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MLCK phosphorylates a protein called myosin regulatory light chain (RLC), which is a part of the myosin molecules composing the thick filament. The activity of MLCK is regulated by the binding of the protein calmodulin (CaM), which occurs when CaM binds Ca^{2+} released from the sarcoplasmic reticulum. Binding of the Ca^{2+}/CaM complex triggers a rearrangement of MLCK in which an autoinhibitory region moves away from the active site for RLC phosphorylation. RLC phosphorylation, which has been shown to vary with the frequency and strength of contractions, results in repositioning of the myosin heads slightly closer to actin binding sites, thereby facilitating actin–myosin interactions.

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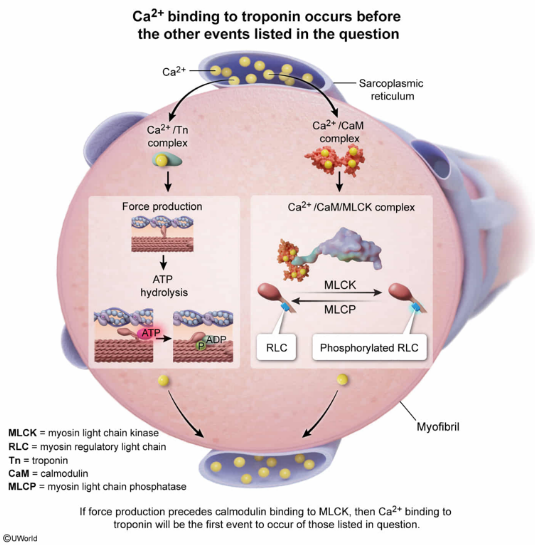
Group	Condition (before or after sensitizing contraction)	Relative force
Wild-type	Before	1.0
	After	2.1
skMLCK KO	Before	0.9
	After	1.1

If skeletal muscle force development precedes the CaM complex binding to MLCK as described in the passage, which of the following events would occur first after stimulation?

- A. Phosphorylation of the RLC of myosin [24%]
- ✔ B. Binding of Ca^{2+} to troponin [42%]
- C. Sarcoplasmic reticulum reuptake of Ca^{2+} released in response to t-tubule depolarization [19%]
- D. Hydrolysis of ATP by myosin [12%]

Correct
42% answered correctly

Explanation:



Muscle force development is initiated by a series of events that occurs in response to muscle fiber membrane depolarization. This series begins with detection of depolarization by a receptor located in sarcolemmal tubes (called **t-tubules**) that penetrate deep into the **muscle fiber**. In response to this depolarization, Ca^{2+} is released from the **sarcoplasmic reticulum** (SR) through a Ca^{2+} channel in the SR membrane. The resultant rise in cytosolic Ca^{2+} leads to binding of Ca^{2+} to a specific type of troponin molecule, and this troponin molecule in turn enables stronger binding of the myosin head to actin. This stronger binding enables myosin to pull on the actin filament, thereby generating tension.

The passage describes the modulation of skeletal muscle force production by skeletal muscle MLCK activation in response to Ca^{2+} release, and the question asks which of the listed events would occur first if skeletal muscle force development precedes CaM binding to MLCK. **If force development precedes CaM binding** to the MLCK, then the Ca^{2+}-mediated events leading to muscle contraction must have already occurred. The **first of these** events among the answer choices **is binding of Ca^{2+} to troponin**.

(Choice A) The phosphorylation of the RLC of myosin described in the passage would occur *after* troponin binding to Ca^{2+}.

(Choice C) Reuptake of Ca^{2+} released in response to t-tubule depolarization is necessary for relaxation to occur *after* a contraction. Therefore, Ca^{2+} reuptake would occur *after* the binding of troponin to Ca^{2+} that is necessary for contraction to occur.

(Choice D) Myosin hydrolysis of ATP occurs after ATP binding to the myosin head, which triggers release of actin from the myosin head after the power stroke. Therefore, myosin hydrolysis would occur *after* troponin binds Ca^{2+} in response to a depolarizing stimulus.

Educational objective:

When Ca^{2+} binds to troponin, troponin causes a slight rearrangement of the components of the thin filament, thereby allowing the stronger binding between actin and myosin that is necessary for skeletal muscle contraction to occur.

Time spent:0 sec
Subject:
Biology

QID:404183
System:
Musculoskeletal System

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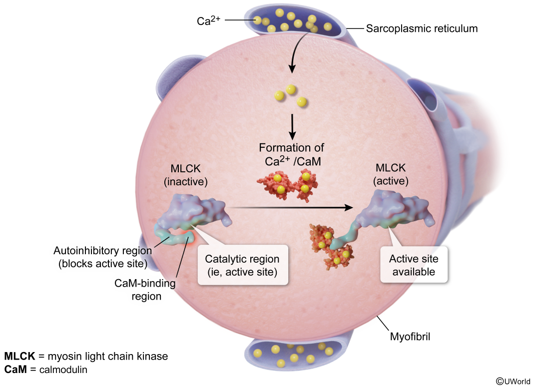
Based on the passage, Ca^{2+} most likely initiates potentiation by directly:

- A. binding troponin to promote movement of tropomyosin on the thin filament. [22%]
- B. binding troponin to prevent movement of tropomyosin on the thin filament. [2%]
- ✔ C. binding CaM to promote intramolecular movement of MLCK. [71%]
- D. binding CaM to prevent intramolecular movement of MLCK. [3%]

Correct
71% answered correctly

Explanation:

Ca^{2+} binding to calmodulin leads to intramolecular movement within MLCK



All muscles produce force in response to elevation of cytosolic Ca^{2+} concentration, although the process differs between **striated** and **smooth muscles**. In striated muscles (ie, **cardiac** and **skeletal muscles**), force production is generated in a series of steps that includes Ca^{2+} binding to a specific **troponin** molecule. This binding eventually leads to the myosin head pulling upon actin and thereby generating tension in the muscle. In smooth muscle, Ca^{2+} binding to CaM activates MLCK to phosphorylate myosin, which in turn causes the force-generating interaction between actin and myosin.

Although not essential for contraction to occur in striated muscles, myosin phosphorylation by MLCK can modulate contractile performance of cardiac and skeletal muscle fibers. The passage describes the phenomenon of force potentiation by MLCK in skeletal muscles, and the question asks how Ca^{2+} initiates the potentiation described. The passage states that Ca^{2+} binding to CaM initiates a rearrangement of MLCK that allows it to bind and phosphorylate the myosin RLC. Therefore, **Ca^{2+}** most likely **initiates potentiation** by directly **binding CaM** to **promote intramolecular movement** of MLCK.

(Choice A) Direct binding of Ca^{2+} to troponin to promote movement of tropomyosin on the thin filament is an essential step for muscle contraction to occur; however, it is *not* part of the mechanism by which MLCK potentiates force production.

(Choice B) Binding of Ca^{2+} to troponin promotes (*not* prevents) movement of tropomyosin on the thin filament.

(Choice D) Binding of Ca^{2+} to CaM promotes (*not* prevents) intramolecular movement of MLCK.

Educational objective:

Elevation of intracellular Ca^{2+} is essential for contraction to occur in all types of muscle.

Time spent:0 sec
Subject:
Biology

QID:404184
System:
Musculoskeletal System

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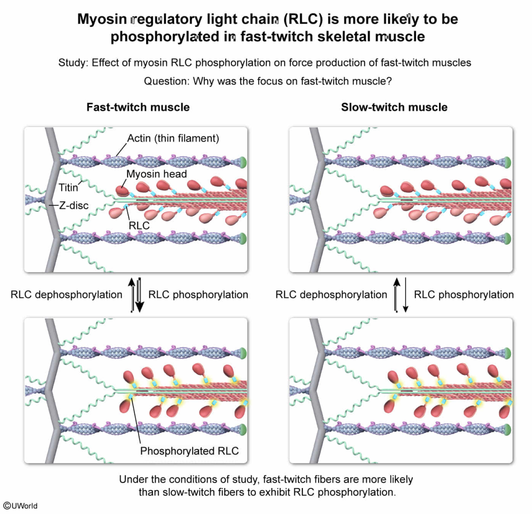
The researchers who carried out the study described in the passage chose to examine the effects of MLCK activation on force production in fast-twitch rather than slow-twitch muscles. What characteristic of MLCK activation is a likely reason for this experimental choice? Potentiation of skeletal muscle force production by MLCK in fast-twitch muscles is:

- A. dependent on higher rates of phosphorylation and lower rates of dephosphorylation than are present in slow-twitch muscles. [58%]
- B. aided by lower CaM concentrations than in slow-twitch muscles. [11%]
- C. easier to detect than in more quickly fatigued slow-twitch muscles. [15%]
- D. facilitated by the presence of higher myoglobin levels than in slow-twitch muscles. [15%]

Correct
58% answered correctly

Explanation:



Skeletal muscle cells contain strands composed of multiple sarcomeres linked in series. Many such strands are contained within one muscle cell, and the muscle cell is typically referred to as a **muscle fiber** due to its elongated structure.

Skeletal muscle fibers can be classified in numerous ways, but the broadest classification scheme organizes muscle fibers according to how quickly they contract. **Slow-twitch fibers** contract more slowly than **fast-twitch fibers** and typically have more mitochondria, capillaries, and myoglobin. This specialization for oxygen delivery, binding, and utilization during aerobic energy production is thought to contribute to the generally greater fatigue resistance of slow-twitch fibers as compared to fast-twitch fibers.

The passage describes research into the effects of MLCK on skeletal muscle force production, and the question asks why the researchers likely focused on fast-twitch rather than slow-twitch muscles. As described in the passage, MLCK phosphorylates the RLC of myosin, and **RLC phosphorylation causes potentiation** of force production. The degree of phosphorylation of a molecule is the net result of the processes of phosphorylation and dephosphorylation. Therefore, it is **likely that** fast-twitch muscles were chosen because the **balance between** RLC **phosphorylation and dephosphorylation in fast-twitch** muscles **favors phosphorylation under** the **conditions of** the **study**.

(Choice B) No information is given to suggest that CaM concentration differs between slow- and fast-twitch muscles. However, even if such a difference exists, a *lower* CaM concentration in fast-twitch muscles would likely limit (*not* aid) MLCK activation.

(Choice C) Fast-twitch muscles are generally *more* quickly fatigued than slow-twitch muscles.

(Choice D) Fast-twitch muscles have lower (*not* higher) myoglobin levels than slow-twitch muscles.

Educational objective:
Fast-twitch muscles contract faster, have fewer mitochondria, and have less myoglobin than slow-twitch muscles.